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A systematic review of behavioral public policy research: origins, mechanisms and outcomes

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Abstract

Purpose – This study aims to identify the origins, mechanisms and outcomes of applying behavioral insight in public policy research.

Design/methodology/approach – The authors conducted a systematic literature review to answer three research questions. The authors identified 387 primary studies, dated from January 2000 to April 2021 and coded them through a thematic analysis. Related studies were obtained through searching in Emerald, ScienceDirect, Sage, Springer, Wiley and Routledge.

Findings – The results identified eight themes for origins, 16 themes for mechanisms/techniques and 13 outcome-related themes. Through the thematic analysis, the major mechanisms of behavioral approach were found to be social marketing, information provision, social norms, incentives, affect, regulation design, framing, salience, defaults, simplification, networking, environment design, scheduled announcements, commitments, attitude-preference-behavior manifestation and combining behavioral and nonbehavioral mechanisms.

Practical implications – The findings of this review help policymakers to design or redesign policy elements.

Originality/value – This review provides the first systematic exploration of the existing literature on behavioral public policy.

Keywords Public policy, Behavioral insight, Nudge, Systematic review

Paper type Literature review

1. Introduction

A fundamental question for policymakers is how to encourage to citizens to adopt a desired behavior. One solution is to apply behavioral insights into the policymaking process (Boun My and Ouvrard, 2019). Some governments have carried out behavioral policies and "nudges," the most prominent subtype of behavioral public policy interventions, keeping choice freedom and taking people in certain directions. In fact, behavioral science has become essential to policy choices in the areas such as environmental protection, tax policy, health care, poverty, consumer protection, retirement and much more (Strassheim, 2020a).

Behavioral instruments apply insights from psychology, behavioral sciences, behavioral economics and neurosciences across a wide range of policy areas. Some scholars have revealed that behavioral public policy is far-reaching than most of the existing applications

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of behavioral insights and nudges (Ewert, 2020). Hence, behavioral public policy includes all modes and means of public policy aiming at affecting citizens' behavior. In the past two decades, there was observed a considerable increase in behavioral research and interventions in the context of public policy on a worldwide scale (Ledderer *et al.*, 2020; Strassheim, 2019; Fayyazi-Azad *et al.*, 2016).

Policy journals are continuously publishing more empirical research than before, and the interest in behavioral public policy and nudge research seems to attract significant interest in the multidisciplinary journals. Previous studies have argued that behavioral public policy can challenge some key principles and conceptualizations in the public policy setting. This study aims to determine the origins, mechanisms and consequences of behavioral public policy in a 20-year related literature with a systematic review.

While there are a lot of published papers, there is no recent systematic review to address the issues. Reviews to date addressing the origins, mechanisms, as well as outcomes of behavioral public policy have been nonsystematic and fragmented. The current review aimed to systematically identify, appraise and synthesize the evidence on the origins, mechanisms and outcomes of behavioral public policy in the recent literature. Hence, this systematic review would contribute to the theory and practice in the context of public policy.

These paper argumentations unfold in three steps. First, it explores the behavioral insight into public policy that has driven nudge techniques. In the second step, the methodology and research process are precisely discussed. Finally, there is a discussion of findings mentioned to academics and practitioners.

2. Theoretical background

The majority of policy tools encompass behavioral assumptions as their basis in an attempt to get people to perform tasks they may not otherwise perform or prepare people to perform tasks that they may not do otherwise (Giesta and Mukherjee, 2018). An overview of behavioral insights literature reveals that the varieties of behavioral approach to public policy are complex and multifaceted. It is considered as a low-cost and easy intervention to change behavior, which concentrates primarily on the choice architecture in different settings of human behavior while maintaining the range of options for choice (Matjasko *et al.*, 2016).

Behavioral insights contain a broad range of instruments in which they may be integrated with or advise traditional types of interventions (Tyers *et al.*, 2019). Recently, behavioral insights are being applied in diverse policy issues such as tax collection, savings for pensions, healthy eating and energy conservation. Policymakers have growingly acknowledged that human behavior is fundamental to making intended policies. In addition, policies mostly influence citizens' behavior, and citizens' behavior would ascertain the success of these policies (Nasrolahi Vosta and Jalilvand, 2022; Dessart and van Bavel, 2017). The basic assumption of behavioral public policy is that strategies of participation and democratization should be informed by insights about the citizens' neurological and cognitive system (Strassheim, 2020b).

Behavioral policies may be considered to be allied to the theory of public choice, which legitimatizes the doctrines of the new public management (John, 2019). Behavioral policies include a myriad of possible change strategies-particular measures used to shape the behaviors of citizens (e.g. feedback, chunking, defaults, social norms, increasing salience, simplification, etc.). Strassheim (2020b) revealed that behavioral public policy clarifies the multiple ways postdemocratic practices both redefine the role of political participation, citizenship and democratic governance and achieve global authority. The wheel of behavior change connects the types of compliance problems (e.g. leakages in capacity and motivation)

with the mechanisms of policy compliance and policy categories (e.g. incentivization or coercion for motivation and training for capacity). It supplies a wide spectrum of policy instruments, such as regulations, financial measures and guidelines (Olejniczak *et al.*, 2020).

Initial applications of behavioral sciences to public policy mostly counted on the nudge approach, i.e. the way options are provided to the citizens (Hagmann *et al.*, 2018). The nudge theory merges psychology and economics to explain the inclinations of decision-making and is recognized for its high relevance to policy development, particularly when the nudges are effective enough to induce libertarian paternalism to facilitate and sustain the desired behavioral changes (Hortal, 2020). This means no to less need for bans and penalties, and of huge expenditures for regulatory actions in policy implementation to get the desired change, as the target population is motivated enough to work for the targeted change to occur. The theory argues that the features of an environment with which an individual makes decisions have influence over him/her in finalizing his/her decisions (Beshears and Kosowsky, 2020).

Nudges contain choice architectures that affect the psychology of the individuals in which they are guided in decision-making, cognizant of the consequences. The nudges are characterized to be not imposing commands and restrictions on the options and, more especially, not to change the financial incentives associated with these options (Kwan *et al.*, 2020). Specific tactics and techniques are used to maximize the potential of nudges to affect an individual to carry out a desired behavior (Beshears and Kosowsky, 2020). As a result, a nudge is a policy intervention for redirecting an agent's choices by very slightly altering their choice conditions (Thaler and Sunstein, 2009). It contrasts with traditional public policies, which usually rely on commands, bans or manipulations of choice incentives (Mongin and Cozic, 2018). Oliver (2019, p. 147) argued that nudges are the dominant framework in behavioral public policy to date. Nudging seems to be firmly positioned in evidence-based policy rhetoric and encourages the use of randomized control trials to determine the effectiveness of a policy (Einfeld, 2019, p. 509).

John and Stoker (2019) set out a new concept called "nudge plus," which adapts their earlier contrast between nudge and the deliberative "think." Nudge Plus is an intervention that has a reflective strategy embedded into a nudge design. It is delivered either as a one-part device in which the nudge and the reflective plus are intrinsically combined or two-part whereby the nudge is extrinsically combined with a deliberative instrument that prompts individual reflection on the nudge (Banerjee and John, 2021). Examples are a GPS device combined with AI technology assistants (one-part) and the dual self-pledge involving multiple commitment contracts catering to an individual's short- and long-term preferences (two-part). Therefore, the nudge plus can be seen as a modification of the toolkit of behavioral public policy.

In summary, incentivized behavioral policies, particularly the nudges and the nudge plus, modify citizens' heuristics and cognitive biases in an attempt to improve the effectiveness of an incentive-modifying intervention or an incentive-structure available at the intended environment (Dede, 2019). However, Loewenstein and Chater (2017) asserted that the behavioral approach to public policymaking must not be considered similar to nudging, as the majority of existing behavioral-inspired policies are not examples of nudges but rather of incentivized behavioral policies.

The oldest work referenced as part of behavioral theory is Simon (1957). Curiously Simon's own account points toward much earlier origins for the behavioral approach and the concept of bounded rationality. Herbert Simon has been considered as a towering figure in the development of behavioral economics but one whose influence by cognitive science and information processing theory has not had as much influence as the heuristics and biases program in contemporary research (Camerer, 2007). Harold Lasswell (1951), Simon's teacher, championed the "policy sciences" as a new field of study for the advancement of

society and the realization of greater human dignity, particularly toward democratic ideals. For Lasswell, the conduct of the policy sciences consisted of the integration of policy analysis and the policy process. Policy process research provided the contextualization of a problem and policy analysis provided the problem-oriented advice. Simon described Lasswell as "the leading behavioralist of them all" (Simon, 1993, p. 49).

Against the rationalist logic, an incrementalist logic has been posited, memorably depicted by Charles Lindblom (1959) as "Muddling Through." This has sparked a longstanding academic debate about the rationalist versus incrementalist nature of policy design (Lindblom, 1959). In this debate, the rationalist view has been critiqued on several of its implicit and explicit assumptions, including its beliefs in: the possibility of synoptic analysis and radical policy change; the solely instrumental role of evidence; the fixed nature of knowledge; the unboundedly rational nature of the policy process; the hegemony of causal knowledge; and the appropriateness of a command-and-control policy delivery model (Lindblom, 1959).

Within the empiricists/experimentalists, which originated from the field of psychology, the early work on heuristics and biases by Tversky and Kahneman (1974) made no reference to Simon work. This could be due to the distinction that they make between reasoning and intuitive thoughts, the latter being more important to judgments and choices. In Kahneman's Nobel lecture, he describes his work with Tversky as an exploration of the psychology of intuitive beliefs and choices and an examination of their bounded rationality (Kahneman, 2003). In addition, Kahneman *et al.* (1982) cite three origins from the 1950s for the heuristics and biases program. These include Paul Meehl's comparisons of clinical and statistical prediction, Ward Edwards' studies of subjective probability and the "investigation of heuristics and strategies of reasoning, for which Herbert Simon offered a program and Jerome Bruner an example."

3. Design of research

3.1 Research questions and research method

The purpose of the current research is to answer three questions about behavioral public policy research. Thus, a systematic literature review is an appropriate approach. In line with the purpose of the current study, the research questions focus on the origins, mechanism and outcomes of behavioral public policy. Table 1 illustrates all the questions of the current review.

3.2 Process of search

To increase the precision of the current systematic review, the searching and selecting processes should be clear. Thus, the next sections specify the processes of data resources selection, the practical strategy to create the strings of search and characterizing the exclusion and inclusion criteria. We pursue the search process developed by Petersen *et al.* (2008). In such

No.	Description
RQ1	What are the main origins of behavioral public policy in the relevant literature? (Why is there a tendency to behavioral public policy?)
RQ2	What are the most common mechanisms of behavioral approach in public policy mentioned in the relevant literature?
RQ3	What are the main outcomes of behavioral public policy in the relevant literature?

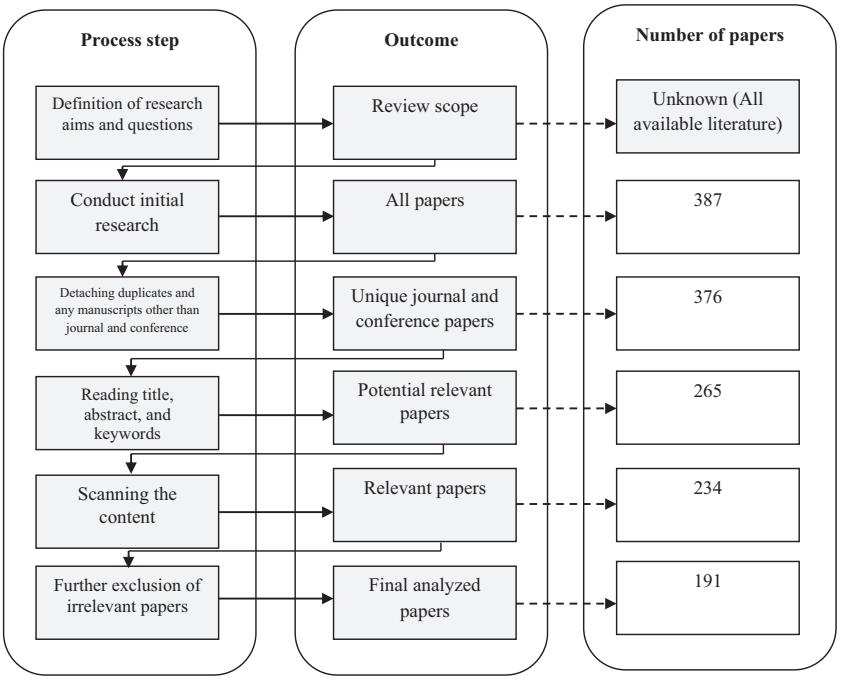
Table 1.
Research questions

Source: Authors' own creation

a process, each stage has a result. Figure 1 shows the complete systematic review process used in this research.

3.2.1 Resources of data and search strategy. Electronic searches were conducted in the following publishers and databases: ScienceDirect, Sage, Emerald, Springer, Wiley and Routledge. The digital databases were picked out as they are important and common databases publishing top journals in the area of public policy. It should be noted that a general WoS search was not used instead of these databases, as some essential journals in the area of public policy or related issues are published in the mentioned databases and have not yet been included in WoS. Further, book chapters, dissertations and theses were excluded because the number of search outputs will become too high. Further, original papers should be published in scientific conferences and journals in the selected databases. The strings of search were generated applying the strategy proposed by Barbosa and Alves (2011): first, the primary keywords were defined; second, the previous known papers in the research area were checked; third, we sought alternative forms of the keywords; fourth, Boolean operator was applied to synthesize them into search strings. The final strings of search are as follows:

> "Nudge" AND "Behavioral insight" or "Behavioral public policy" or "Behavioral insight" AND "Public policy"



Source: Authors' own creation

Figure 1.
Process of search

These search strings were used to search within all components of the papers, including keywords, abstract, title and main body. The process of search started in April 2021. Several sample papers then went through, and the synonyms were affirmed. To narrow the number of search outputs and to have more precise outputs from the search in the selected databases, each part of the search string was put in quotation marks to discover the exact phrase.

3.2.2 Inclusion and exclusion criteria. The criteria are used to include relevant and exclude irrelevant research (Petersen *et al.*, 2008). This assures that relevant papers would be analyzed. In this research, the following criteria for inclusion and exclusion of the papers were applied (Table 2). Book chapters, dissertations and theses were excluded because the number of search outputs will become too high; therefore, it will be infeasible to analyze all of the outputs. Additionally, the original paper should be published in scientific conferences and journals. Hence, the search scope was minimized to journal and conference papers. Zotero, the reference management software, was used to manage the papers and remove duplicate references. In doing so, all the initial 387 papers were sorted first based on the name of the author and then by going through the column of the author's name; if the name of the author recurred in sequence, then the titles and the years were checked. In case of duplication of any paper that was not downloaded from the six target publishers, including Emerald, Springer, Sage, Routledge, ScienceDirect and Wiley, it was removed. In the final step, we had unique articles to analyze by thematic analysis. Thematic analysis allows for the explicit identification of outstanding themes and provides structured and organized ways of dealing with the target literature under the themes. It was made flexible, providing the reviewers with significant scope and as a means of integrating both quantitative and qualitative types of manuscript (Abdullah and Harding, 2020).

4. Results

In this section, the review results have been reported according to the three questions of research. The results are based on the analysis of 191 selected papers.

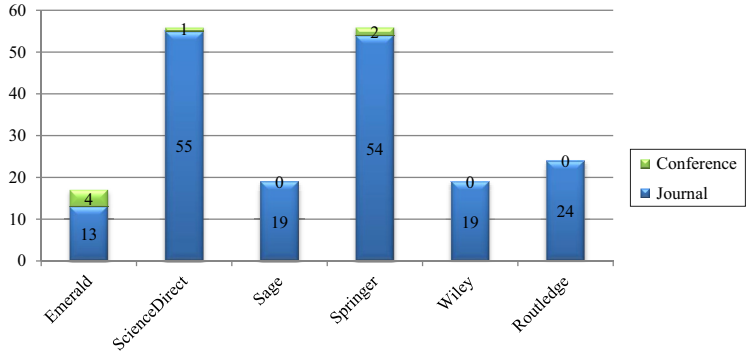
4.1 Demographic data

An overview of the reviewed papers was provided with respect to the database and publication type. The distribution of publication fora was identified for 191 unique papers. Figure 2 illustrates the distribution of the major databases and the publication types. The distribution of publications shows the important databases for this research area and further research. Figure 2 depicts the publication distribution categorized by resource

Inclusion	Exclusion
Papers implying on "Nudges" and/or "Behavioral Public Policy"	Papers that are outside of "Nudges" and/or "Behavioral Public Policy" domain
Papers in English language	Papers in other languages
Conference and journal papers	Theses, dissertations, editorial notes, book chapters, presentations, work reports, newsletters, grey literature and trade literature
Papers that are peer reviewed	Papers that are nonpeer reviewed
Published between 2000 and 2021	Duplicate papers

Table 2. Criteria for exclusion and inclusion

Source: Authors' own creation



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Figure 2. Distribution of database and publication type

containing conference proceedings and journal papers. All the resources have been provided in Appendix. Accordingly, ScienceDirect and Springer have published 58.64% (56 papers for each publisher) of the resources in the area of behavioral public policy, followed by Routledge (12.57%), Wiley (9.95%), Sage (9.95%) and Emerald (8.9%). In addition, the journals were found to be the prime publication venue for behavioral public policy research (96.33%).

4.2 Origins of behavioral public policy (RQ1)

To answer *RQ1* (Why is there a tendency to behavioral public policy?), 191 papers were analyzed by thematic analysis. We categorized the codes extracted from the literature into eight different themes: the development of more perfect public policies, achievement of long-term outcomes, understanding the cognitive mechanisms behind decision-making, application of more effective and flexible policy instruments, complexity of policy problems and environment, networking and interactions of sciences/experts, steering citizens toward making better choices and increasing popularity and interest in behavioral policies. Description of codes associated with each theme has been provided in Table 3. The resource/s of each code is presented in Table 3.

4.3 Mechanisms of behavioral public policy (RQ2)

To provide answers to *RQ2*, the final 191 articles were analyzed to reveal the most common mechanisms of behavioral approach in public policy. The results showed that there are 16 themes for the mechanisms of behavioral approach in public policy. They include social marketing, information provision, social norms, incentives, affect, regulation design, framing, salience, defaults, simplification, networking, environment design, scheduled announcements, commitments, attitude-preference-behavior manifestation and combining behavioral and nonbehavioral mechanisms. The extracted codes of each mechanism theme can be found in Table 4.

4.4 Outcomes of behavioral public policy (RQ3)

In an attempt to answer *RQ3*, the authors have analyzed 191 final papers, as provided in Appendix, published in periods of 20 years from 2000 to the beginning of 2021. The results indicate that there are several important outcomes for the implementations of behavioral public policy (see Table 5). Twelve themes related to the behavioral policies outcomes were identified. They include enhancement of social participation in the policy implementation, higher levels of effectiveness and efficiency, improvement of public services and goods, knowledgeable citizens, knowledgeable policymakers, social welfare growth, helping to

Themes	Codes	Reference/s
Development of more perfect public policies	To be more effective than top-down policy	Colasanti *et al.* (2020)
	To shift from government to governance	Pillay and van den Bergh (2016)
	Risks associated with public policymaking	Brooks (2015)
	Using behavioral insights to improve policies in areas (such as health care, savings, finance, discrimination, employment and the environment)	Andreadakis (2020)
	To design better policies that are less intrusive and more cost effective than many existing policies	Van Gestel *et al.* (2018), Kooreman and Prast (2010), Rathi and Chunekar (2015), Croson and Treich (2014), Viscusi (2019)
	Risk-averse policymakers needing to predictable ways by the outcomes of their previous decisions	Brada *et al.* (2015)
	Influencing decisions made with less top-down, conscious control	Felsen and Reiner (2015)
	To make better policy recommendations	Shogren *et al.* (2010)
	To exploit additional policy dimensions other than financial incentives	Shogren *et al.* (2010)
Achievement of long-term outcomes	Long-term changes require long-term action	Sinclair (2015)
	Interventions improve decision-making outcomes	Babcock *et al.* (2012), Frerichs (2011)
	Easily implementation	Rathi and Chunekar (2015)
Understanding the cognitive mechanisms behind decision-making	A welcome change from the questionable assumptions about rationality and decision-making	MacKay and Quigley (2018)
	To counteract a variety of cognitive biases and misconceptions clouding mental models that may not always maximize the expected utility	Ramya and Baral (2019), Brada *et al.* (2015), Fabbri and Faure (2018), Felsen and Reiner (2015), Hagman *et al.* (2015), Hepburn *et al.* (2010), Peeters and Schuilenburg (2017), Schnellenbach and Schubert (2019), Battaglio *et al.* (2018), Galizzi (2014)
	Understanding the links between cognitive and social mechanisms	Strassheim (2019)
	Changing preferences through social mechanisms	Brennan (2018)
	To give policymakers a richer model of human behavior	DellaValle and Sareen (2020)
	Targeting individuals' intuitive way of thinking	Hagman *et al.* (2015)
	To understand the shortcuts and heuristics that people use to make decisions	Peeters and Schuilenburg (2017)
	To understand the mechanism of change underlying the policy initiative	Gopalan and Pirog (2017)

Table 3.
Origins of behavioral public policy

(*continued*)

Themes	Codes	Reference/s
Application of more effective and flexible policy instrument	Existing instruments have very limited systematic behavioral input	Giesta and Mukherjee (2018)
	Adapting systematic biases in the traditional policy instruments	Garcia-Sierra *et al.* (2015)
	Existing instruments have to be tailored toward behavioral patterns of citizens	Giesta and Mukherjee (2018)
	Behavioral mechanisms enrich the way policy instruments are mixed and set-up	Giesta and Mukherjee (2018)
	To improve the efficacy of the classical tools	Bauer and Reisch (2019)
	To offer new policy tools that are less intrusive and more flexible than bans and taxes	Bauer and Reisch (2019)
	Calling for a test-learn-adapt approach, using experiments, running pilot tests and improving policies with an empirical and iterative process	Bauer and Reisch (2019)
	Pilot-testing of policy initiatives using randomized control trials to increase the internal validity of causal impacts	Gopalan and Pirog (2017)
Complexity of policy problems and environment	To inform policymaking by providing evidence about policy problems and the expected behavioral implications of policy tools	Ewert (2020)
	Increasing complexity and ambiguity of legal rules	Pillay and van den Bergh (2016)
	Complex and multidimensional choice environment	Smith and Toprakkiran (2018)
	Complex problems involving behavioral and structural dimensions	Ewert (2020)
	Different dynamics influencing the problem space and the solution space	Jones (2017)
	Considering the grand challenges of society	Cappa *et al.* (2020)
Networking and interactions of sciences/ experts	Expanding beyond social sciences to hard sciences like neuroscience	Alm and Sheffrin (2016)
	Rational choice challenged by the library and empirical works	Brennan (2018)
	Emergence of cognitive and behavioral enhancement technologies	Lehoux *et al.* (2017)
	Increasing interlinkages between global networks of expertise	Strassheim (2020)
	Global interactions	Strassheim (2020)
To steer citizens toward making better choices	The necessity of behavioral change to tackle societal problems	Tummers (2019)
	To facilitate and encourage the citizens in making better, desired and optimized choices	Evers *et al.* (2018), Fabbri and Faure (2018)
	To change individual, collective and organizational behaviors	Ewert (2020)

(*continued*)

Table 3.

Themes	Codes	Reference/s
	To direct individuals toward better behavior for personal and societal benefits	Cappa *et al.* (2020), Junghans *et al.* (2015), Oliver (2019), Rachlin (2015)
	To encourage citizens to voluntarily contribute to the public goods	Czajkowski *et al.* (2019)
	To support individual behavior and to promote collective action	DellaValle and Sareen (2020)
	To move individuals to action in a meaningful way	Garcíaa and Vila (2020)
	To alter choice environments to improve selections	Vecchio and Cavallo (2019)
	To use the biases to direct behavior toward better options	Bauer and Reisch (2019)
Increasing popularity and interest in behavioral policies	Massive interest in the use of nudges to improve the quality of public administration	John (2019)
	Well-accepted nudge	Van Gestel *et al.* (2018)
	New insights from psychology and behavioral economics	Moseley and Stoker (2013)
	Significant interest among political commentators is "behavior change" using "environmental design" as opposed to legislation	Seymour and Vlaev (2012)
	Widespread adoption and increasingly popularity as a regulatory tool	Hill (2018), Leeman *et al.* (2015)

Table 3. **Source:** Authors' own creation

sustainability, higher savings and tax revenues, higher donations and charities, more perfect regulations and compliance, better returns for the educational system, designing better policies to solve problems and directing citizens in a better way.

4.5 Threats to validity

In this research type, potential problems of bias should be identified, and it is necessary to allow readers to inquire the results credibility and validity. In fact, there were several threats to the validity of this research. Hence, it is important to assess the validity of the current review study. Three kinds of threats need to be considered to validate the results credibility: internal validity, external validity and construct validity. Internal validity refers to the extent to which the research design and implementation reduce biased results or systematic error. On the other hand, external validity refers to the extent to which the research results may be used to other settings outside of the research. Replication of review research is a way to lessen the threats of external and internal validity. Moreover, using a well-defined methodology declined the bias of this research. The external validity was ensured as we made no generalizations, claims and projections.

Construct validity alludes to what extend the inferences may be made with respect to the questions of research. Construct validity evaluates the operability of a construct against theory. To ensure construct validity of the current study, the research questions were defined in the steps of the research method. Thus, it can provide the same interpretations for other researchers who are interested in iterating a similar research process in the future. To detect the maximum number of relevant papers for systematic review associated with the

Themes	Codes	Reference/s
Social marketing	Assessing returns on investment and social impacts beyond traditional cost-benefit analysis	DeCicca *et al.* (2017), Dessart and van Bavel (2017), Galizzi *et al.* (2018)
	Social marketing leveraging behavioral insights to improve the social image of alternatives	Dessart and van Bavel (2017), French (2011), Garcia-Sierra *et al.* (2015), Matjasko *et al.* (2016), Chriss (2015), Metcalfe *et al.* (2020)
	Intersecting strategies of social marketing	Mulderrig (2017), Crawshaw (2013)
Information provision	Providing customized or social comparative feedback	Gaker and Walker (2011), Dogruel (2019), Ledderer *et al.* (2020), Cappa *et al.* (2020), Crago *et al.* (2020), Rathi and Chunekar (2015), Castaldi *et al.* (2020), Yoong *et al.* (2020)
	Providing rapid, accurate, reliable and understandable information	Gaker and Walker (2011), Dogruel (2019), Ledderer *et al.* (2020), Cappa *et al.* (2020), Crago *et al.* (2020), Garcia-Sierra *et al.* (2015), Howarth *et al.* (2019), Jones (2017), Myers and Souza (2020), Charlier and Kirakozian (2019), Harbers *et al.* (2020), Galizzi (2014), Guthrie *et al.* (2015)
	Organizing various informational/ media campaigns	Jagau and Vyrastekova (2017), Elbel *et al.* (2014), Ledderer *et al.* (2020), Agarwal *et al.* (2017), Broers *et al.* (2019), Crago *et al.* (2020)
	Labeling to provide standard information	Cioffi *et al.* (2015), Andreadakis (2020)
	Contextualizing information	Crago *et al.* (2020)
Social norms	Promoting and activation of social norms	Gaker and Walker (2011), Just and Swigert (2016), Martin (2015), Crago *et al.* (2020), Czajkowski *et al.* (2019), Drews *et al.* (2020), Malone *et al.* (2017), Matjasko *et al.* (2016), Rathi and Chunekar (2015), Valatin *et al.* (2016), Andreadakis (2020), Bicchieri and Dimant (2019), Rocha *et al.* (2020)
	Social comparisons	Boun My and Ouvrard (2019), Myers and Souza (2020), Charlier and Kirakozian (2019), Pogarsky and Herman (2019)
Incentives	Using incentive-compatible experimental methods	Galizzi *et al.* (2018)
	Incentive design	Drews *et al.* (2020), Jensen *et al.* (2020), Seymour and Vlaev (2012), Valatin *et al.* (2016), Charlier and Kirakozian (2019), Rocha *et al.* (2020), Junghans *et al.* (2015), Galizzi (2014)
	Providing prosocial appeals	Jensen *et al.* (2020), Harbers *et al.* (2020), Pogarsky and Herman (2019)
	Offering multiple incentives at intervals	Vlaev *et al.* (2019)
	Personalization	Valatin *et al.* (2016)

(*continued*)

Table 4. Mechanisms of behavioral public policy

Themes	Codes	Reference/s
Affect	Social context	Alm and Sheffrin (2016)
	Social emotions linking to the behavior	Jagau and Vyrastekova (2017)
	Shifts in feelings	Hobbs (2017)
	Social pressure favoring alternatives and behaviors	Hobbs (2017), Garcia-Sierra *et al.* (2015)
Combining behavioral and nonbehavioral mechanisms	Combination of personalized tools and public nudges	Schröder and Lyon (2013)
	Combination of nudging and mandatory measures	Ali *et al.* (2014)
	To combine behaviorally informed policy tools with existing tools such as taxes and subsidies	Gopalan and Pirog (2017)
Regulation design	Using rule-intermediaries	Talesh and Pélisse (2019), Kendall (2010), Bell (2011), Brooks (2015), Ali *et al.* (2014), Carroll *et al.* (2018), Jones (2017), Matjasko *et al.* (2016), Hill (2018), MacKay and Quigley (2018), Rischkowsky and Döring (2008)
	Justifying regulations to correct for cognitive biases	Brennan (2018)
Attitude-preference-behavior manifestation	Making the required action a habit by using recurrent cues and prompts	Seymour and Vlaev (2012)
	Attitude change through cognitive dissonance	Hobbs (2017)
	Stated preference approach	Czajkowski *et al.* (2019)
Framing	Framing public issues by the media	Kenterelidou (2012)
	Framed text messaging	Castleman and Page (2015), Garcia-Sierra *et al.* (2015), Nakamura *et al.* (2017), Gopalan and Pirog (2017), Guthrie *et al.* (2015)
	Designing menus to cause framing effects	Matjasko *et al.* (2016)
Salience	Moral education to increase the moral salience	Martin (2015), Andreadakis (2020)
	Increasing the salience of messaging	Malone *et al.* (2017), Rocha *et al.* (2020)
	Presenting choices in a more salient or interesting light	Junghans *et al.* (2015), Yoong *et al.* (2020)
Defaults	Setting smart, automatic and thoughtfully default options	Leigh (2015), Coffinoa *et al.* (2020), Drews *et al.* (2020), Garcíaa and Vila (2020), Garcia-Sierra *et al.* (2015), Malone *et al.* (2017), Matjasko *et al.* (2016), Momsen and Stoerk (2014), Rathi and Chunekar (2015), Schulz *et al.* (2018), Zarghamee *et al.* (2016), Rocha *et al.* (2020), SP19, Kooreman and Prast (2010), Metcalfe *et al.* (2020), Yoong *et al.* (2020), Gopalan and Pirog (2017)
	Manifestation of status quo bias	Garcia-Sierra *et al.* (2015), Rathi and Chunekar (2015), Zarghamee *et al.* (2016)
	To change the default rules	Andreadakis (2020)

(*continued*)

Table 4.

Themes	Codes	Reference/s
Simplification	Small and simple behavioral interventions	Strassheim (2019)
	Simplifying decision-making	Giesta and Mukherjee (2018)
	Placement/convenience	Metcalfe *et al.* (2020)
	Identifying and removing actual or perceived barriers/friction	Valatin *et al.* (2016)
Networking	Collaborative nudging	Whitehead *et al.* (2020)
	Using social networks	Valatin *et al.* (2016)
	Implementing training program	Ramya and Baral (2019), Dogruel (2019), Hobbs (2017), Garcia-Sierra *et al.* (2015)
Environment design	Interventions in the choice environment	Drews *et al.* (2020)
	Redesigning the environment when people make choices among complex (multiattribute) alternatives	Seymour and Vlaev (2012)
Scheduled announcements	Announcement interventions exploiting social image	Barron and Nurminen (2020)
	Timing interventions/treatments at critical points	Myers and Souza (2020), Valatin *et al.* (2016), Andreadakis (2020)
	Reminders	Gillitzer and Sinning (2020), Andreadakis (2020), Namazu *et al.* (2018), Yoong *et al.* (2020), Gopalan and Pirog (2017)
	Warnings	Andreadakis (2020)
Commitments	Making public commitment using related devices	Myers and Souza (2020), Rathi and Chunekar (2015), Valatin *et al.* (2016), Kooreman and Prast (2010)
	Goal setting	Rathi and Chunekar (2015)
	Precommitment	Pogarsky and Herman (2019)

Source: Authors' own creation

Table 4.

research questions, the search queries were defined and refined based on the achieved results and adopted synonym keywords in the search strings and conducted the queries in the most well-known electronic databases. As another facet of construct validity, it was assured that all the relevant papers are included. Therefore, we tried to compound as many papers as possible by using the electronic databases. However, we might have lost several papers that electronic databases could not detect.

5. Discussion

Analysis of the microfoundations of public decision-making shows that public managers are faced with numerous and insurmountable complex impacts. Behavior-altering techniques provide a better understanding of how policymakers can affect decisions through behavioral insight that nudges public managers in ways that lead to outcomes that are more suitable (Battaglio *et al.*, 2018). This research is a systematic review of literature that seeks to respond to three major questions:

Q1. Why is there a tendency to behavioral public policy? (Origins)

Q2. What are the most common mechanisms of behavioral approach in public policy mentioned in the relevant literature?

Q3. What are the main outcomes of behavioral public policy in the relevant literature?

Themes	Codes	Reference/s
Enhancement of social participation in the policy implementation	Helping target interventions to improve motivation to attend appointments	Aggarwal *et al.* (2016)
	Encouraging individual responsibility and altruism	Aggarwal *et al.* (2016)
	Managing behaviors in a better way	Brooks (2015)
	Engaging citizens in public policies	John (2019)
	Shaping social action	Kenterelidou (2012)
	Contribute to transforming individuals in their role as participants in the game of politics	Schubert (2017)
	To increase participation in a given program	Clark *et al.* (2014)
Higher levels of efficiency and effectiveness	Lower administrative costs	Aggarwal *et al.* (2016)
	Effectiveness fed into future program planning	French (2011)
	Enhancement of public policies efficiency	John (2019)
	Cheaply and effectively behavioral change	Beshears and Kosowsky (2020), Crawshaw (2013)
	To improve the effectiveness of interventions relying on social information	Bicchieri and Dimant (2019)
Improvement of public goods and services	To improve the performances of markets and governments	Congleton (2019)
	To improve the delivery of governmental services	Gopalan and Pirog (2017)
	To improve compliance and use of government services by the public	Gopalan and Pirog (2017)
Knowledgeable citizens	Promotion and enhancement of knowledge circulation	Colasanti *et al.* (2020)
	To contribute to citizen science	Cappa *et al.* (2020)
	To induce individuals to better understand themselves and use that knowledge to improve their own lives	Schubert (2017)
Knowledgeable policymakers	Helps legitimate and instantiate neoliberal political rationalities	Mulderrig (2017)
	Creating more precise knowledge on how actual behavior deviates from full rationality	Kooreman and Prast (2010)
	Improving psychosocial outcomes	Trudel-Fitzgerald *et al.* (2019)
	Providing explanations for the inherent complexity of economic life	Kendall (2010)
Social welfare growth	To steer individuals toward choices or behaviors that benefit the target individual and society	Junghans *et al.* (2015), Bicchieri and Dimant (2019), Byerly *et al.* (2018)
	Assisting healthier and nutritional food choices	Schröder and Lyon (2013), Coffinoa *et al.* (2020), Fennis *et al.* (2020), Pitts *et al.* (2016), Bauer and Reisch (2019), Byerly *et al.* (2018), Davidson *et al.* (2020)
	To reduce cigarette consumption improves social welfare	DeCicca *et al.* (2017)

Table 5.
Outcomes of behavioral public policy

(*continued*)

Themes	Codes	Reference/s
	Promoting societal well-being	Dessart and van Bavel (2017)
	Choosing outcomes leading to better health and lifestyle choices	Brooks (2015)
	Changes in lifestyle and the achievement of healthy lifestyles	Hobbs (2017), Crawshaw (2013)
	Prevention and control of smoking	Kenterelidou (2012)
	Welfare-maximizing	Elbel *et al.* (2014)
	Improving the supply of healthy food	Ewert (2020)
	Poverty alleviation	Ramani *et al.* (2017)
	To enhance energy justice	DellaValle and Sareen (2020)
Helping to sustainability	Reducing and preventing waste	Jagau and Vyrastekova (2017), Rivers *et al.* (2017), Zhang and Wang (2020)
	Environmentally responsible decision-making	Ramya and Baral (2019)
	Reinforcing public concerns for climate issues	Pillay and van den Bergh (2016)
	Reduction in energy consumption	Agarwal *et al.* (2017), Crago *et al.* (2020)
	To improve the awareness about environmentally-related issues	Cappa *et al.* (2020)
	Reducing energy use	Agarwal *et al.* (2017), Crago *et al.* (2020)
	Promise for sustainability	Byerly *et al.* (2018), Kaplan *et al.* (2018)
Higher savings and tax revenues	To drive engagement and improve financial outcomes for consumers	Tyers *et al.* (2019)
	Increasing number of people saving for old age	Sinclair (2015)
	To increase long-term saving patterns	Garcíaa and Vila (2020)
	Better tax collection	John (2019)
	Effectively raising taxpayer's compliance	Fonseca and Grimshaw (2017)
Higher donations and charities	To double the fraction of donors and the revenue for charities	Schulz *et al.* (2018)
	Facilitating charitable giving	Hobbs (2017), John (2019)
More perfect regulations and compliance	Affecting, controlling or monitoring relations between rule-makers and rule-takers	Meyer and Rosinger (2019)
	Constructing the content and meaning of law	Talesh and Pélisse (2019)
	To boost court attendance	Chivers and Barnes (2018)
	Making more ethical decisions by regulators	Cai (2018)
Better returns for educational system	Increased access of young people to education opportunities	John (2019)
	To increase college entry among populations	Castleman and Page (2015)
	To achieve meaningful improvements in students' educational outcomes	Castleman and Page (2015)
	Affecting students' postsecondary engagement in supplementing structural changes	Meyer and Rosinger (2019)

(*continued*)

Table 5.

Themes	Codes	Reference/s
Designing better policies to solve problems	Encouraging policymakers to communicate more clearly with clients and customers	Room (2016)
	Responding to problems without challenging existing social, political and economic relationships	Smith and Toprakkiran (2018)
	Improving the ability to predict behavior and to devise policies	Alm and Sheffrin (2016)
	Understanding of the cognitive, social and even moral factors driving decision-making	Moseley and Stoker (2013)
	The highest acceptance among both nationals and expatriates	Lambert and Lee (2018)
Directing citizens in a better way	To encourage prosocial behavior	Nagatsu (2015)
	Simplifying the choice process	Elbel *et al.* (2014)
	Providing a focal point for coordination	Barron and Nurminen (2020)
	Making the preferred choice	Felsen and Reiner (2015)

Table 5. **Source:** Authors' own creation

In this research, after defining the strategy of research, we searched through the adopted databases and detected 387 papers. All of them were not unique and relevant; thus, the inclusion and exclusion criteria were defined. Then we had 191 unique and relevant papers to analyze. To analyze the 191 selected papers, thematic analysis was used. Regarding the origins of behavioral public policy, we extracted eight themes:

(1) the development of more perfect public policies;
(2) the achievement of long-term outcomes;
(3) understanding the cognitive mechanisms behind decision-making;
(4) the application of more effective and flexible policy instruments;
(5) the complexity of policy problems and environment;
(6) networking and interactions of sciences/experts;
(7) steering citizens toward making better choices; and
(8) increasing popularity and interest in behavioral policies.

Regarding the mechanisms of behavioral public policy, 16 mechanisms of behavioral approach implementation were identified, i.e. social marketing, information provision, social norms, incentives, affect, regulation design, framing, salience, defaults, simplification, networking, environment design, scheduled announcements, commitments, attitude-preference-behavior manifestation and combining behavioral and nonbehavioral mechanisms.

Finally, we recognized 13 outcome-related themes, i.e. enhancement of social participation in the policy implementation, higher levels of effectiveness and efficiency, improvement of public services and goods, knowledgeable citizens, knowledgeable policymakers, social welfare growth, helping to sustainability, higher savings and tax revenues, higher donations and charities, more perfect regulations and compliance, better returns for the educational system, designing better policies to solve problems and directing citizens in a better way.

6. Implications

Behavioral public policy is based on the assumption that strategies of democratization and participation need to be informed by insights about the cognitive and neurological system of citizens. This article focused on the origins, mechanisms and consequences of behavioral approaches for governance and participation. There has been a heated debate on the normative underpinnings and democratic consequences of behavioral public policy. Critics have argued that nudges and other behavioral interventions are a form of "stealth politics" that undermine citizen's freedom by means of manipulation and that they should be rejected because of the multiple ways they contravene democratic values such as fairness, transparency and empowerment. Behavioral public policy is based on the capability of actors to combine both epistemic and political authority, i.e. expectations toward their capacity to produce and validate knowledge about governance that is deemed scientifically sound as well as politically relevant and legitimate.

This review provided an implementation framework, including 16 mechanisms/techniques of behavioral policy, to change citizens' behavior, which is easy to communicate but simultaneously acknowledges mechanisms in a nonlinear, combinatory way. For practitioners, that helps to design or redesign change mechanisms and test the effectiveness of different configurations of policy elements. For researchers, it offers a canvas for reconstructing assumptions on causal mechanisms. The study aimed at identifying and systematizing different change strategies that share the same change logic. This is especially relevant to the dynamically growing practice of nudges and behavioral approaches that uses a spectrum of heuristics and biases.

Behavioral public policies can be considered as a panacea in terms of providing a low-cost and effective policy instrument. They are influential largely because they have adapted a public choice conception of action – citizens are rational actors, but their choices are framed. Hence, the reorientation of framing can be viewed as effective at producing policy change without major interventions either in terms of finance or bureaucracy. As a result, behavioral policies fit well with governance in times of austerity. However, this approach ignores two fundamental issues. First, citizens are rational but not always in a utility maximizing way. They are able to reflect upon, and vary, their responses in relation to the architecture of choice. Second, the choice architecture in some areas may be much broader than behavioralist suggestion. Behavioralism focuses on the immediate and short-term choice architecture of actors and not the broader social, economic and political environment within which decisions are made. Consequently, any behavioral policy changes for major social problems are likely to have less effect that advocates claim.

Further, behavioral public policy can be seen as the result of the capability of actors to claim both political and epistemic authority by mobilizing experimental evidence and by redefining rationality, autonomy and authenticity. As behavioral insights become more influential in policy design and implementation, it is necessary to ensure that the insights are derived from studies that are sensitive to the policy context and appropriate for the population to which they are applied.

6.1 Research limitations

This systematic review has encountered several limitations. First, searches have been limited to only six scientific databases. Hence, we could not consider all the conference and journal databases. Further, we assumed only peer-reviewed papers, such as conference and journal papers, and we included no magazine or book section in our research. In the realm of search keywords, we attempted to use synonym words as much as possible to obtain the maximum number of results. For example, we applied "public policy and behavioral insight" as the alternatives for "behavioral public policy." It is possible that we lose several papers that have applied other terms.

References

Abdullah, R. and Harding, R. (2020), "Preferences and experiences of Muslim patients and their families in Muslim-majority countries for end-of-life care: a systematic review and thematic analysis", *Journal of Pain and Symptom Management*, Vol. 60 No. 6, pp. 1223-1238.

Banerjee, S. and John, P. (2021), "Nudge plus: incorporating reflection into behavioral public policy", *Behavioural Public Policy*, pp. 1-16.

Barbosa, O. and Alves, C. (2011), "A systematic mapping study on software ecosystems", Workshop on Software Ecosystems, Brussels, pp. 15-26.

Battaglio, R.P., Belardinelli, P., Bellé, N. and Cantarelli, P. (2018), "Behavioral public administration ad fontes: a synthesis of research on bounded rationality, cognitive biases, and nudging in public organizations", *Public Administration Review*, Vol. 79 No. 3, pp. 304-320.

Beshears, J. and Kosowsky, H. (2020), "Nudging: progress to date and future directions", *Organizational Behavior and Human Decision Processes*, Vol. 161, pp. 3-19.

Boun My, K. and Ouvrard, B. (2019), "Nudge and tax in an environmental public goods experiment: Does environmental sensitivity matter?", *Resource and Energy Economics*, Vol. 55, pp. 24-48.

Camerer, C. (2007), "Behavioral economics", California Institute of Technology, Working Paper.

Dede, O.C. (2019), "Behavioral policies and inequities: the case of incentivized smoking cessation policies", *Journal of Economic Methodology*, Vol. 26 No. 3, pp. 272-289.

Dessart, F.J. and van Bavel, R. (2017), "Two converging paths: behavioural sciences and social marketing for better policies", *Journal of Social Marketing*, Vol. 7 No. 4, pp. 355-365.

Einfeld, C. (2019), "Nudge and evidence based policy: fertile ground", *Evidence and Policy*, Vol. 15 No. 4, pp. 509-524.

Ewert, B. (2020), "Moving beyond the obsession with nudging individual behaviour: towards a broader understanding of behavioural public policy", *Public Policy and Administration*, Vol. 35 No. 3, pp. 337-360.

Fayyazi-Azad, A., Rahmati, M.H. and Khannifar, H. (2016), "Designing a model for factors affecting social responsibility in Iran's national oil company", *Social Capital Management*, Vol. 4 No. 2, pp. 181-197.

Giesta, S. and Mukherjee, I. (2018), "Behavioral instruments in renewable energy and the role of big data: a policy perspective", *Energy Policy*, Vol. 123, pp. 360-366.

Hagmann, D., Siegrist, M. and Hartmann, C. (2018), "Taxes, labels, or nudges? Public acceptance of various interventions designed to reduce sugar intake", *Food Policy*, Vol. 79, pp. 156-165.

Hortal, A. (2020), "Evidence-based policies, nudge theory and Nancy Cartwright: a search for causal principles", Behavioral Public Policy, pp. 1-20.

John, P. (2019), "The international appeal of behavioural public policy: is nudge an Anglo-American phenomenon?", *Journal of Chinese Governance*, Vol. 4 No. 2, pp. 144-162.

John, P. and Stoker, G. (2019), "Rethinking the role of experts and expertise in behavioural public policy", *Policy and Politics*, Vol. 47 No. 2, pp. 209-225.

Kahneman, D. (2003), "Maps of bounded rationality: psychology for behavioral economics", *American Economic Review*, Vol. 93 No. 5, pp. 1449-1475.

Kahneman, D., Slovic, P. and Tversky, A. (1982), *Judgment under Uncertainty: Heuristics and Biases*, Cambridge University Press, Cambridge.

Kwan, Y.H., Cheng, T.Y., Yoon, S., Ho, L.Y.C., Huang, C.W., Chew, E.H., Thumboo, J., Østbye, T. and Low, L.L. (2020), "A systematic review of nudge theories and strategies used to influence adult health behaviour and outcome in diabetes management", *Diabetes and Metabolism*, Vol. 46 No. 6, pp. 450-460.

Lasswell, H.D. (1951), "The policy orientation", in Lerner, D. and Lasswell, H.D. (Eds), *The Policy Sciences: Recent Developments in Scope and Methods*, Stanford University Press, Stanford, CA.

Ledderer, L., Kjær, M., Madsen, E.K., Busch, J. and Fage-Butler, A. (2020), "Nudging in public health lifestyle interventions: a systematic literature review and metasynthesis", *Health Education and Behavior*, Vol. 47 No. 5, pp. 749-764.

Lindblom, C. (1959), "The science of muddling through", *Public Administration Review*, Vol. 19 No. 2, pp. 79-88.

Loewenstein, G. and Chater, N. (2017), "Putting nudges in perspective", *Behavioural Public Policy*, Vol. 1 No. 1, pp. 26-53.

Matjasko, J.L., Cawley, J.H., Baker-Goering, M.M. and Yokum, D.V. (2016), "Applying behavioral economics to public health policy: illustrative examples and promising directions", *American Journal of Preventive Medicine*, Vol. 50 No. 5, pp. 13-19.

Mongin, P. and Cozic, M. (2018), "Rethinking nudge: not one but three concepts", *Behavioural Public Policy*, Vol. 2 No. 1, pp. 107-124.

Nasrolahi Vosta, L. and Jalilvand, M.R. (2022), "Electronic trust-building for hotel websites: a social exchange theory perspective", *Journal of Islamic Marketing*, doi: 10.1108/JIMA-05-2022-0119.

Olejniczak, K., Ś liwowski, P. and Leeuw, F. (2020), "Comparing behavioral assumptions of policy tools: framework for policy designers", *Journal of Comparative Policy Analysis: Research and Practice*, Vol. 22 No. 6, pp. 498-520.

Oliver, A. (2019), *Reciprocity and the Art of Behavioural Public Policy*, Cambridge University Press.

Petersen, K., Feldt, R., Mujtaba, S. and Mattsson, M. (2008), "Systematic mapping studies in software engineering", *Proceedings of 12th International Conference on Evaluation and Assessment in Software Engineering (EASE)*, *University of Bari, Bari, Italy*.

Simon, H. (1957), *Models of Man*, Wiley, New York, NY.

Simon, H.A. (1993), "The state of American political science: professor Lowi's view of our discipline", *PS: Political Science and Politics*, Vol. 26 No. 1, pp. 1-7.

Strassheim, H. (2019), "Behavioural mechanisms and public policy design: preventing failures in behavioural public policy", *Public Policy and Administration*, Vol. 36 No. 2, doi: 10.1177/0952076719827062.

Strassheim, H. (2020a), "The rise and spread of behavioral public policy: an opportunity for critical research and self-reflection", *International Review of Public Policy*, Vol. 2 No. 1, pp. 115-128.

Strassheim, H. (2020b), "De-biasing democracy, behavioural public policy and the post-democratic turn", *Democratization*, Vol. 27 No. 3, pp. 461-476.

Thaler, R.H. and Sunstein, C. (2009), *Nudge*, Penguin Books, New York, NY.

Tversky, A. and Kahneman, D. (1974), "Judgement under uncertainty: heuristics and biases", *Science*, Vol. 185 No. 4157, pp. 1124-1131.

Tyers, R., Sweeney, M. and Moon, B. (2019), "Harnessing behavioural insights to encourage consumer engagement in the British energy market: results from a field trial", *Journal of Behavioral and Experimental Economics*, Vol. 80, pp. 162-176.

Appendix

Paper code	Details: author/s, year, title, journal, volume, number, pages
E1	Aggarwal, A., Davies, J. and Sullivan, R. (2016), "'Nudge' and the epidemic of missed appointments: Can behavioural policies provide a solution for missed appointments in the health service?", *Journal of Health Organization and Management*, Vol. 30 No. 4, pp. 558-564
E2	Bevir, M. and Needham, C. (2017), "Decentring social policy: narratives, resistance, and practices", *International Journal of Sociology and Social Policy*, Vol. 37 No. 11–12, pp. 626-638
E3	Colasanti, N., Fiori, V. and Frondizi, R. (2020), "Promoting knowledge circulation in public libraries: the role of gamification", *Library Management*, Vol. 41 No. 8/9, pp. 669-676
E4	DeCicca, P., Kenkel, D., Liu, F. and Wang, H. (2017), "Behavioral Welfare Economics and FDA Tobacco Regulations", *Human Capital and Health Behavior* (*Advances in Health Economics and Health Services Research*, Vol. 25), Emerald Publishing Limited, pp. 143-179
E5	Dessart, F.J. and van Bavel, R. (2017), "Two converging paths: behavioural sciences and social marketing for better policies", *Journal of Social Marketing*, Vol. 7 No. 4, pp. 355-365
E6	French, J. (2011), "Why nudging is not enough", *Journal of Social Marketing*, Vol. 1 Iss 2, pp. 154-162
E7	Gaker, D. and Walker, J.L. (2011), "Insights on car-use behaviors from behavioral economics", Lucas, K., Blumenberg, E. and Weinberger, R. (Ed.) Auto Motives, Emerald Publishing Limited, pp. 107-120
E8	Galizzi, M.M., Harrison, G.W. and Miraldo, M. (2018), "Experimental methods and behavioral insights in health economics: estimating risk and time preferences in health", *Health Econometrics* (*Contributions to Economic Analysis*, Vol. 294), Emerald Publishing Limited, pp. 1-21
E9	Jagau, H.L. and Vyrastekova, J. (2017), "Behavioral approach to food waste: an experiment", *British Food Journal*, Vol. 119 No. 4, pp. 882-894
E10	Just, D.R. and Swigert, J.M. (2016), "The role of nudges in reducing food waste", *Food Security in a Food Abundant World* (*Frontiers of Economics and Globalization*, Vol. 16), Emerald Publishing Limited, pp. 215-224
E11	Ramani, S.V., Parihar, R. and Sen, S. (2017), "On nudging MNE toward SDG1: a policy perspective", *International Business and Management* (*International Business and Management*, Vol. 33), Emerald Publishing Limited, pp. 89-129
E12	Ramya, S.M. and Baral, R. (2019), "'Mind' matters! A conceptual framework using mental models and green nudging to drive corporate environmental responsibility", *Management Decision*, Vol. ahead-of-print No. ahead-of-print
E13	Schröder, M. and Lyon, P. (2013), "Embedding healthy eating: nudging or toolbox?", Nutrition and Food Science, Vol. 43 No. 4, pp. 330-338
E14	Sinclair, D. (2015), "The coalition government and ageing policy: a critique", *Quality in Ageing and Older Adults*, Vol. 16 No. 1, pp. 3-13
E15	Talesh, S. and Pélisse, J. (2019), "How Legal Intermediaries Facilitate or Inhibit Social Change", *Studies in Law, Politics, and Society* (*Studies in Law, Politics, and Society*, Vol. 79), Emerald Publishing Limited, pp. 111-145
E16	Pillay, C. and van den Bergh, J. (2016), "Human health impacts of climate change as a catalyst for public engagement: combining medical, economic and behavioural insights", *International Journal of Climate Change Strategies and Management*, Vol. 8 No. 5, pp. 578-596
E17	Kendall, K. (2010), "Nudge – improving decisions about health, wealth and happiness", *Sustainability Accounting, Management and Policy Journal*, Vol. 1 No. 1, pp. 106-112.
R1	Altman, M. (2005), "Behavioral economics, power, rational inefficiencies, fuzzy sets, and public policy", *Journal of Economic Issues*, 39:3, 683-706
R2	Brada, J. C., Kubíček, J., Kutan, A. M. and Tomšik, V. (2015), "Inflation targeting: insights from behavioral economics", *Eastern European Economics*, Vol. 53 No. 5, pp. 357-376

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Table A1.
Resources

Paper code	Details: author/s, year, title, journal, volume, number, pages
R3	Bell, K. (2011), "Legislating abjection? Secondhand smoke, tobacco control policy and the public's health", *Critical Public Health*, Vol. 21 No. 1, pp. 49-62
R4	Brennan, T. R. (2018), "The rise of behavioral economics in regulatory policy: rational choice or cognitive limitation?", *International Journal of the Economics of Business*, Vol. 25 No. 1, pp. 97-108
R5	Brooks, T. (2015), "Alcohol and controlling risks through nudges", *The New Bioethics*, Vol. 21 No. 1, pp. 46-55
R6	Dede, O. C. (2019), "Behavioral policies and inequities: the case of incentivized smoking cessation policies", *Journal of Economic Methodology*, Vol. 26 No. 3, pp. 272-289
R7	Dogruel, L. (2019), "Privacy nudges as policy interventions: comparing US and German media users' evaluation of information privacy nudges", *Information, Communication and Society*, Vol. 22 No. 8, pp. 1080-1095
R8	Hobbs, J. (2017), "Nudging charitable giving: the ethics of nudge in international poverty reduction", *Ethics and Global Politics*, Vol. 10 No. 1, pp. 37-57
R9	Houk, T., DiSilvestro, R. and Jensen, M. (2016), "Smoke and mirrors: subverting rationality, positive freedom, and their relevance to nudging and/or smoking policies", *The American Journal of Bioethics*, Vol. 16 No. 7, pp. 20-22
R10	John, P. (2019), "The international appeal of behavioural public policy: is nudge an Anglo-American phenomenon?", *Journal of Chinese Governance*, Vol. 4 No. 2, pp. 144-162
R11	Kenterelidou, C. (2012), "Framing public health issues: the case of smoking ban in Greece, public health policy framing equals healthy framing of public policy?", *Journal of Communication in Healthcare*, Vol. 5 No. 2, pp. 116-128
R12	Lee, C. and Ma, L. (2019), "The role of policy labs in policy experiment and knowledge transfer: a comparison across the UK, Denmark, and Singapore", *Journal of Comparative Policy Analysis: Research and Practice*, doi: 10.1080/13876988.2019.1668657
R13	Lehoux, P., Williams-Jones, B., Grimard, D. and Proulx, S. (2017), "Technologies of the self in public health: insights from public deliberations on cognitive and behavioural enhancement", *Critical Public Health*, Vol. 27 No. 3, pp. 373-383
R14	Martin, C. (2015), "Nudging the public sphere: a Habermasian perspective on public deliberation as an aim of moral education", *Journal of Moral Education*, Vol. 44 No. 4, pp. 440-456
R15	Mulderrig, J. (2017), "Multimodal strategies of emotional governance: a critical analysis of 'nudge' tactics in health policy", *Critical Discourse Studies*, Vol. 15 No. 1, pp. 39-67
R16	Necker, S. (2016), "Why do scientists cheat? Insights from behavioral economics", *Review of Social Economy*, Vol. 74 No. 1, pp. 98-108
R17	Olejniczak, K., Ś liwowski, P. and Leeuw, F. (2020), "Comparing behavioral assumptions of policy tools: framework for policy designers", *Journal of Comparative Policy Analysis: Research and Practice*, Vol. 22 No. 6, pp. 498-520
R18	Room, G. (2016), "Nudge or nuzzle? Improving decisions about active citizenship", *Policy Studies*, Vol. 37 No. 2, pp. 113-128
R19	Smith, M. and Toprakkiran, N. (2018), "Behavioural insights, nudge and the choice environment in obesity policy", *Policy Studies*, Vol. 40 No. 2, pp. 173-187
R20	Spiegel, M., Hill, M. and Gennetian, L. A. (2020), "Harnessing a behavioral economic framework for supporting providers in improving early childhood care", *Early Years*, doi: 10.1080/09575146.2020.1732877
R21	Strassheim, H. (2020), "De-biasing democracy. Behavioural public policy and the post-democratic turn", *Democratization*, Vol. 27 No. 3, pp. 461-476
R22	Turner, K., Alabi, O. and Race, J. (2020), "Nudging policymakers: a case study of the role and influence of academic policy analysis", *Journal of European Public Policy*, Vol. 27 No. 8, pp. 1270-1286
R23	Van Gestel, L. C., Kroese, F. M. and De Ridder, D. T. D. (2018), "Nudging at the checkout counter – a longitudinal study of the effect of a food repositioning nudge on healthy food choice", *Psychology and Health*, Vol. 33 No. 6, pp. 800-809

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Table A1.

Paper code	Details: author/s, year, title, journal, volume, number, pages
R24	Yeung, R. and Nguyen-Hoang, P. (2020), "Using texting to nudge urban public school students to and through college", *Journal of Research on Technology in Education*, Vol. 52 No. 1, pp. 113-127
S1	Ali, P., Ramsay, I. and Read, C. (2014), "Behavioural law and economics: regulatory reform of consumer credit and consumer financial services", *Common Law World Review*, Vol. 43, pp. 298-343
S2	Alm, J. and Sheffrin, S. M. (2016), Using behavioral economics in public economics, *Public Finance Review*, Vol. 45 No. 1, pp. 4-9
S3	Cantarelli, P., Belle, N. and Belardinelli, P. (2020), "Behavioral public HR: experimental evidence on cognitive biases and debiasing interventions", *Review of Public Personnel Administration*, Vol. 40 No. 1, pp. 56-81
S4	Congdon, W. J. and Shankar, M. (2018), "The role of behavioral economics in evidence-based policymaking", *The ANNALS of the American Academy of Political and Social Science*, Vol. 678 No. 1, pp. 81-92
S5	Crawshaw, P. (2013), "Public health policy and the behavioural turn: the case of social marketing", *Critical Social Policy*, Vol. 33 No. 4, pp. 616-637
S6	Elbel, B., Gillespie, C. and Raven, M. C. (2014), "Presenting quality data to vulnerable groups: charts, summaries or behavioral economic nudges?" *Journal of Health Services Research and Policy*, Vol. 19 No. 3, pp. 161-168
S7	Ewert, B. (2020), "Moving beyond the obsession with nudging individual behaviour: towards a broader understanding of behavioural public policy", *Public Policy and Administration*, Vol. 35 No. 3, pp. 337-360
S8	Fonseca, M. A. and Grimshaw, S. B. (2017), "Do behavioral nudges in pre-populated tax forms affect compliance? Experimental evidence with real taxpayers", *Journal of Public Policy and Marketing*, Vol. 36 No. 2, pp. 213-226
S9	Jones, R., Pykett, J. and Whitehead, M. (2014), "The geographies of policy translation: how nudge became the default policy option", *Environment and Planning C: Government and Policy*, Vol. 32, pp. 54-69
S10	Kasdan, D. O. (2020), "Do Koreans like being nudged? Survey evidence for the contextuality of behavioral public policy", *Rationality and Society*, Vol. 32 No. 3, pp. 313-333
S11	Le Clainche, C. and Tubeuf, S. (2016), "Nudging, intervening or rewarding: a discussion on the constraints and the degree of control on health status", *Politics, Philosophy and Economics*, Vol. 15 No. 2, pp. 170-189
S12	Ledderer, L., Kjær, M., Madsen, E. K., Busch, J. and Fage-Butler, A. (2020), "Nudging in public health lifestyle interventions: a systematic literature review and metasynthesis", *Health Education and Behavior*, Vol. 47 No. 5, pp. 749-764
S13	Leigh, A. (2015), "How behavioural economics does and can shape public policy", *The Economic and Labour Relations Review*, Vol. 26 No. 2, pp. 339-346
S14	Mukherjee, I. and Giest, S. (2020), Behavioural insights teams (BITs) and policy change: an exploration of impact, location, and temporality of policy advice, *Administration and Society*, Vol. 52 No. 10, pp. 1538-1561
S15	Pickett, J. T. (2018), "Using behavioral economics to advance deterrence research and improve crime policy: some illustrative experiments", *Crime and Delinquency*, Vol. 64 No. 12, pp. 1636-1659
S16	Poole, M. K., Mundorf, A. R., Englar, N. K. and Rose, D. (2019), "Advancing public health policy in Louisiana through evidence-based strategies: successes and opportunities", *Health Promotion Practice*, Vol. 20 No. 1, pp. 128-134
S17	Schaap, N. T. W. and van de Riet, O. A. W. T. (2012), "Behavioral insights model: overarching framework for applying behavioral insights in transport policy analysis" *Transportation Research Record*, Vol. 2322 No. 1, pp. 42-50
S18	Strassheim, H. (2019), "Behavioural mechanisms and public policy design: preventing failures in behavioural public policy", *Public Policy and Administration*, doi: https://doi.org/10.1177/0952076719827062

Table A1. (*continued*)

Paper code	Details: author/s, year, title, journal, volume, number, pages
S19	Whitehead, M., Jones, R. and Pykett, J. (2020), "Questioning post-political perspectives on the psychological state: behavioural public policy in the Netherlands", *Environment and Planning C: Politics and Space*, Vol. 38 No. 2, pp. 214-232
SD1	Agarwal, S., Rengarajan, S., Sing, T. F. and Yang, Y. (2017), "Nudges from school children and electricity conservation: evidence from the 'Project Carbon Zero' campaign in Singapore", *Energy Economics*, Vol. 61, pp. 29-41
SD2	Barron, K. and Nurminen, T. (2020), "Nudging cooperation in public goods provision", *Journal of Behavioral and Experimental Economics*, Vol. 88, p. 101542
SD3	Bernheim, B. D. and Taubinsky, D. (2018), "Behavioral public economics", *Handbook of Behavioral Economics*, Vol. 1, pp. 381-516
SD4	Beshears, J. and Kosowsky, H. (2020), "Nudging: progress to date and future directions", *Organizational Behavior and Human Decision Processes*, Vol. 161, pp. 3-19
SD5	Boun My, K. and Ouvrard, B. (2019), "Nudge and tax in an environmental public goods experiment: Does environmental sensitivity matter?" *Resource and Energy Economics*, Vol. 55, pp. 24-48
SD6	Broers, V. J. V., Van de Broucke, S., Taverne, C. and Luminet, O. (2019), "Investigating the conditions for the effectiveness of nudging: cue-to-action nudging increases familiar vegetable choice", *Food Quality and Preference*, Vol. 71, pp. 366-374
SD7	Bunten, A., Porter, L., Burgess-Allen, J., Howell-Jones, R., Jackson, J., Ward, D., Staples, V., Staples, P., Rowthorn, H., Saei, A., van Schaik, P., Tydeman, E., Blair, P., Hugueniot, O., Gold, N. and Chadborn, T. (2021), "Using behavioural insights to reduce sugar in primary school children's packed lunches in derby; a cluster randomised controlled trial", *Appetite*, Vol. 157, p. 104987
SD8	Cappa, F., Rosso, F., Giustiniano, L. and Porfiri, M. (2020), "Nudging and citizen science: the effectiveness of feedback in energy-demand management", *Journal of Environmental Management*, Vol. 269, p. 110759
SD9	Carroll, K. A., Samek, K. and Zepeda, L. (2018), "Food bundling as a health nudge: investigating consumer fruit and vegetable selection using behavioral economics", *Appetite*, Vol. 121, pp. 237-248
SD10	Castleman, B. L. and Page, L. C. (2015), "Summer nudging: can personalized text messages and peer mentor outreach increase college going among low-income high school graduates?" *Journal of Economic Behavior and Organization*, Vol. 115, pp. 144-160
SD11	Cioffi, C. E., Levitsky, D. A., Pacanowski, C. R. and Bertz, F. (2015), "A nudge in a healthy direction: the effect of nutrition labels on food purchasing behaviors in university dining facilities", *Appetite*, Vol. 92, pp. 7-14
SD12	Coffinoa, J. A., Udo, T. and Hormes, J. M. (2020), "Nudging while online grocery shopping: a randomized feasibility trial to enhance nutrition in individuals with food insecurity", *Appetite*, Vol. 152, p. 104714
SD13	Crago, C. L., Spraggon, J. M. and Hunter, E. (2020), "Motivating non-rate paying households with feedback and social nudges: a cautionary tale", *Energy Policy*, Vol. 145, p. 111764
SD14	Czajkowski, M., Zagórska, M. and Hanley, N. (2019), "Social norm nudging and preferences for household recycling", *Resource and Energy Economics*, Vol. 58, p. 101110
SD15	DellaValle, N. and Sareen, S. (2020), "Nudging and boosting for equity? Towards a behavioural economics of energy justice", *Energy Research and Social Science*, Vol. 68, p. 101589
SD16	Doberstein, C., Hickey, R. and Li, E. (2016), "Nudging NIMBY: Do positive messages regarding the benefits of increased housing density influence resident stated housing development preferences?" *Land Use Policy*, Vol. 54, pp. 276-289
SD17	Drews, S., Exadaktylos, F. and van den Bergh, J. C. J. M. (2020), "Assessing synergy of incentives and nudges in the energy policy mix", *Energy Policy*, Vol. 144, p. 111605
SD18	Engel, C. and Kurschilgen, M. (2020), "The fragility of a nudge the power of self-set norms to contain a social dilemma", *Journal of Economic Psychology*, Vol. 81, p. 102293

(*continued*)

Table A1.

Paper code	Details: author/s, year, title, journal, volume, number, pages
SD19	Fennis, B. M., Gineikiene, J., Barauskaite, D. and van Koningsbruggen, G. M. (2020), "Nudging health: scarcity cues boost healthy consumption among fast rather than slow strategists (and abundance cues do the opposite)", *Food Quality and Preference*, Vol. 85, p. 103967
SD20	Garcíaa, J. M. and Vila, J. (2020), "Financial literacy is not enough: the role of nudging toward adequate long term saving behavior", *Journal of Business Research*, Vol. 112, pp. 472-477
SD21	Garcia-Sierra, M., van den Bergh, J. C. J. M. and Miralles-Guasch, C. (2015), "Behavioural economics, travel behavior and environmental-transport policy", *Transportation Research Part D*, Vol. 41, pp. 288-305
SD22	Gazheli, A., Antal, M. and van den Bergh, J. (2015), "The behavioral basis of policies fostering long-run transitions: stakeholders", *Limited Rationality and Social Context*, Vol. 69, pp. 14-30
SD23	Giesta, S. and Mukherjee, I. (2018), "Behavioral instruments in renewable energy and the role of big data: a policy perspective", *Energy Policy*, Vol. 123, pp. 360-366
SD24	Gillitzer, G. and Sinning, M. (2020), "Nudging businesses to pay their taxes: Does timing matter?" *Journal of Economic Behavior and Organization*, Vol. 169, pp. 284-300
SD25	Howarth, C., Kantenbacher, J., Guida, K., Roberts, T. and Rohse, M. (2019), "Improving resilience to hot weather in the UK: the role of communication, behaviour and social insights in policy interventions", *Environmental Science and Policy*, Vol. 94, pp. 258-261
SD26	Jensen, C. L., Andersen, L. M., Hansena, L. G. and Henningsen, G. (2020), "Is social nudging too emotionally taxing? A field experiment of public utilities and electricity consumers in Denmark", *Energy Research and Social Science*, Vol. 67, p. 101515
SD27	Pitts, S. B. J., Wu, Q., Sharpe, P. A., Rafferty, A. P., Elbel, B., Ammerman, A. S., Payne, C. R., Hopping, B. N., McGuirt, J. T. and Wall-Bassett, E. D. (2016), "Preferred healthy food nudges, food store environments, and customer dietary practices in 2 low-income southern communities", *Journal of Nutrition Education and Behavior*, Vol. 48 No. 10, pp. 735-742
SD28	Jones, B.D. (2017), "Behavioral rationality as a foundation for public policy studies", *Cognitive Systems Research*, Vol. 43, pp. 63-75
SD29	Kersh, R. (2015), "Of nannies and nudges: the current state of U.S. obesity policymaking", *Public Health*, Vol. 129 No. 8, pp. 1083-1091
SD30	Malone, S. K., Ziporyn, T. and Buttenheim, A. M. (2017), "Applying behavioral insights to delay school start times", *Sleep Health*, Vol. 3 No. 6, pp. 483-485
SD31	Kurz, V. (2018), "Nudging to reduce meat consumption: immediate and persistent effects of an intervention at a university restaurant", *Journal of Environmental Economics and Management*, Vol. 90, pp. 317-341
SD32	Labanc, N. and Bertoldi, P. (2018), "Beyond energy efficiency and individual behaviours: policy insights from social practice theories", *Energy Policy*, Vol. 115, pp. 494-502
SD33	Lambert, L. A. and Lee, J. (2018), "Nudging greywater acceptability in a Muslim country: comparisons of different greywater reuse framings in Qatar", *Environmental Science and Policy*, Vol. 89, pp. 93-99
SD34	Lieberoth, A., Jensen, N. H. and Bredah, T. (2018), "Selective psychological effects of nudging, gamification and rational information in converting commuters from cars to buses: a controlled field experiment", *Transportation Research Part F*, Vol. 55, pp. 246-261
SD35	Matjasko, J. L., Cawley, J. H., Baker-Goering, M. M. and Yokum, D. V. (2016), "Applying behavioral economics to public health policy: illustrative examples and promising directions", *American Journal of Preventive Medicine*, Vol. 50 No. 1, pp. 13-19
SD36	Momsen, K. and Stoerk, T. (2014), "From intention to action: Can nudges help consumers to choose renewable energy?" *Energy Policy*, Vol. 74, pp. 376-382
SD37	Moseley, A. and Stoker, G. (2013), "Nudging citizens? Prospects and pitfalls confronting a new heuristic", *Resources, Conservation and Recycling*, Vol. 79, pp. 4-10
SD38	Myers, E. and Souza, M. (2020), "Social comparison nudges without monetary incentives: evidence from home energy reports", *Journal of Environmental Economics and Management*, Vol. 101, p. 102315

Table A1. (*continued*)

Paper code	Details: author/s, year, title, journal, volume, number, pages
SD39	Osman, M. and Nelson, W. (2019), "How can food futures insight promote change in consumers choices, are behavioural interventions (e.g. nudges) the answer?" *Futures*, Vol. 111, pp. 116-122
SD40	Pugatch, T. and Wilson, N. (2018), "Nudging study habits: a field experiment on peer tutoring in higher education", *Economics of Education Review*, Vol. 62, pp. 151-161
SD41	Rathi, S. S. and Chunekar, A. (2015), "Not to buy or can be 'nudged' to buy? Exploring behavioral interventions for energy policy in India", *Energy Research and Social Science*, Vol. 7, pp. 78-83
SD42	Rivers, N., Harris, S. S. and Young, N. (2017), "Using nudges to reduce waste? The case of Toronto's plastic bag levy", *Journal of Environmental Management*, Vol. 188, pp. 153-162
SD43	Roberto, C. A., Pomeranz, J. L. and Fisher, J. O. (2014), "The need for public policies to promote healthier food consumption: a comment on Wansink and Chandon (2014)", *Journal of Consumer Psychology*, Vol. 24 No. 3, pp. 438-445
SD44	Rowbotham, S., McKinnon, M., Marks, L. and Hawe, P. (2019), "Research on media framing of public policies to prevent chronic disease: a narrative synthesis", *Social Science and Medicine*, Vol. 237, p. 112428
SD45	Schneider, A. L. J. and Graham, T.C. N. (2017), "Nudging and shoving: using in-game cues to guide player exertion in exergames", *Entertainment Computing*, Vol. 19, pp. 83-100
SD46	Schubert, C. (2017), "Green nudges: Do they work? Are they ethical?" *Ecological Economics*, Vol. 132, pp. 329-342
SD47	Schulz, J. F., Thiemann, P. and Thoni, C. (2018), "Nudging generosity: choice architecture and cognitive factors in charitable giving", *Journal of Behavioral and Experimental Economics*, Vol. 74, pp. 139-145
SD48	Seymour, B. and Vlaev, I. (2012), "Can, and should, behavioural neuroscience influence public policy?" *Trends in Cognitive Sciences*, Vol. 16 No. 9, pp. 449-451
SD49	Spandagos, C., Yarime, M., Baarke, E. and Ng, T. L. (2020), "Triple target policy framework to influence household energy behavior: satisfy, strengthen, include", *Applied Energy*, Vol. 269, p. 115117
SD50	Tyers, R., Sweeney, M. and Moon, B. (2019), "Harnessing behavioural insights to encourage consumer engagement in the British energy market: results from a field trial", *Journal of Behavioral and Experimental Economics*, Vol. 80, pp. 162-176
SD51	Valatin, G., Moseley, D. and Dandy, N. (2016), "Insights from behavioural economics for forest economics and environmental policy: potential nudges to encourage woodland creation for climate change mitigation and adaptation?" *Forest Policy and Economics*, Vol. 72, pp. 27-36
SD52	Vandenbroele, J., Slabbinck, H., Van Kerckhove, A. and Vermeir, I. (2019), "Mock meat in the butchery: nudging consumers toward meat substitutes", *Organizational Behavior and Human Decision Processes*, doi: https://doi.org/10.1016/j.obhdp.2019.09.004
SD53	Vecchio, R. and Cavallo, C. (2019), "Increasing healthy food choices through nudges: a systematic review", *Food Quality and Preference*, Vol. 78, p. 103714
SD54	Wolbertus, R., Kroesen, M., van den Hoed, R. and Chorus, C. G. (2018), "Policy effects on charging behaviour of electric vehicle owners and on purchase intentions of prospective owners: natural and stated choice experiments", *Transportation Research Part D*, Vol. 62, pp. 283-297
SD55	Zarghamee, H. S., Messer, K. D., Fooks, J. R., Schulze, W. D., Wu, S. and Yan, J. (2016), Nudging charitable giving: three field experiments, *Journal of Behavioral and Experimental Economics*, Vol. 00, pp. 1-13
SD56	Zhang, Z. and Wang, X. (2020), "Nudging to promote household waste source separation: mechanisms and spillover effects", *Resources, Conservation & Recycling*, Vol. 162, p. 105054
SP1	Amir, O., Ariely, D., Cooke, A., Dunning, D. and Epley, N. (2005), "Psychology, behavioral economics, and public policy", *Marketing Letters*, Vol. 16 No. 3/4, pp. 443-454
SP2	Anagnostopoulou, E., Urbancic, J., Bothos, E., Magoutas, B., Bradesko, L. Schrammel, J. and Mentzas, G. (2018), "From mobility patterns to behavioural change: leveraging travel behaviour and personality profiles to nudge for sustainable transportation", *Journal of Intelligent Information Systems*, Vol. 54, pp. 157-178

(*continued*)

Table A1.

Paper code	Details: author/s, year, title, journal, volume, number, pages
SP3	Andreadakis, Z. (2020), "Public mission under scarcity: behavioral insights into Greek higher education", *Tertiary Education and Management*, Vol. 26, pp. 91-104
SP4	Arno, A. and Thomas, S. (2016), "The efficacy of nudge theory strategies in influencing adult dietary behaviour: a systematic review and meta-analysis", *Arno and Thomas BMC Public Health*, Vol. 16, pp. 1-12
SP5	Häußermann, J. J. (2019), "Nudging and participation: a contractualist approach to behavioural policy", *Philosophy of Management*, Vol. 19, pp. 45-68
SP6	Babcock, L., Congdon, W. J., Katz, L. F. and Mullainathan, S. (2012), "Notes on behavioral economics and labor market policy", *Journal of Labor Policy*, Vol. 1, pp. 1-14
SP7	Bauer, J. M. and Reisch, L. A. (2019), "Behavioural insights and (un)healthy dietary choices: a review of current evidence", *Journal of Consumer Policy*, Vol. 42, pp. 3-45
SP8	Berggren, N. (2012), "Time for behavioral political economy? An analysis of articles in behavioral economics", *The Review of Austrian Economics*, Vol. 25, pp. 199-221
SP9	Bicchieri, C. and Dimant, E. (2019), "Nudging with care: the risks and benefits of social information", *Public Choice*, doi: https://doi.org/10.1007/s11127-019-00684-6
SP10	Castaldi, G., Cecere, G. and Zoli, M. (2020), "'Smoke on the beach': on the use of economic vs behavioral policies to reduce environmental pollution by cigarette littering", *Economia Politica*, doi: https://doi.org/10.1007/s40888-020-00205-5
SP11	Charlier, C. and Kirakozian, A. (2019), Public policies for household recycling when reputation matters, *Journal of Evolutionary Economics*, Vol. 30, pp. 523-557
SP12	Chivers, B. and Barnes, G. (2018), "Sorry, wrong number: tracking court attendance targeting through testing a BNudge text", *Cambridge Journal of Evidence Based Policy*, Vol. 2, pp. 4-34
SP13	Chriss, J. J. (2015), "Nudging and social marketing", *Social Science and Public Policy*, Vol. 52, pp. 54-61
SP14	Congleton, R. D. (2019), "Behavioral economics and the Virginia school of political economy: overlaps and complementarities", *Public Choice*, doi: https://doi.org/10.1007/s11127-019-00679-3
SP15	Croson, R. and Treich, N. (2014), "Behavioral environmental economics: promises and challenges", *Environmental Resource Economics*, Vol. 58, pp. 335-351
SP16	Rocha, C. A. A. and Hunziker, M. H. L. (2020), "Behavior-analytic view on nudges: individual, technique, and ethics", *Behavior and Social Issues*, Vol. 29, pp. 138-161
SP17	Santos, Q. D., PerezCueto, F. J. A., Rodrigues, V. M., Appleton, K., Giboreau, A., Saulais, L., Monteleone, E., Dinnella, C., Brugarolas, M. and Hartwell, H. (2020), "Impact of a nudging intervention and factors associated with vegetable dish choice among European adolescents", *European Journal of Nutrition*, Vol. 59, pp. 231–247
SP18	Evers, C., Marchiori, D. R., Junghans, A. F., Cremers, J. and De Ridder, D. T. D. (2018), "Citizen approval of nudging interventions promoting healthy eating: the role of intrusiveness and trustworthiness", *BMC Public Health*, Vol. 18, p. 1182
SP19	Ewert, B. (2017), "Promoting health in schools: theoretical reflections on the settings approach versus nudge tactics", *Social Theory & Health*, Vol. 15, pp. 430-447
SP20	Fabbri, M. and Faure, M. (2018), "Toward a "constitution" for behavioral policy making", *International Review of Economics*, Vol. 65, pp. 241-270
SP21	Esposito, L. L. M. F. (2017), "A dismal reality: behavioural analysis and consumer policy", *Journal of Consumer Policy*, Vol. 40, pp. 193-216
SP22	Felsen, G. and Reiner, P. B. (2015), "What can neuroscience contribute to the debate over nudging?" *Review of Philosophy and Psychology*, Vol. 6, pp. 469-479
SP23	Trudel-Fitzgerald, C., Millstein, R. A., Hippel, C. V., Howe, C. J., Tomasso, L. P., Wagner, G. R. and VanderWeele, T. J. (2019), "Psychological well-being as part of the public health debate? Insight into dimensions, interventions, and policy", *BMC Public Health*, Vol. 19, p. 1712
SP24	Forberger, S., Reisch, L., Kampfmann, T. and Zeeb, H. (2019), "Nudging to move: a scoping review of the use of choice architecture interventions to promote physical activity in the general population", *International Journal of Behavioral Nutrition and Physical Activity*, Vol. 16, p. 77

Table A1. (*continued*)

Paper code	Details: author/s, year, title, journal, volume, number, pages
SP25	Frerichs, S. (2011), "False promises? A sociological critique of the behavioural turn in law and economics", *Journal of Consumer Policy*, Vol. 34, pp. 289-314
SP26	Garoupa, N. (2003), "Behavioral economic analysis of crime: a critical review", *European Journal of Law and Economics*, Vol. 15, pp. 5-15
SP27	Guala, F. and Mittone, L. (2015), "A political justification of nudging", *Review of Philosophy and Psychology*, Vol. 6, pp. 385-395
SP28	Hagman, W., Andersson, D., Västfjäll, D. and Tinghög, G. (2015), "Public views on policies involving nudges", *Review of Philosophy and Psychology*, Vol. 6, pp. 439-453
SP29	Harbers, M. C., Beulens, J. W. J., Rutters, F., Boer, F., Gillebaart, M., Sluijs, I. and van der Schouw, Y. T. (2020), "The effects of nudges on purchases, food choice, and energy intake or content of purchases in real-life food purchasing environments: a systematic review and evidence synthesis", *Nutrition Journal*, Vol. 19, pp. 1-27
SP30	Hepburn, C., Duncan, S. and Papachristodoulou, A. (2010), "Behavioural economics, hyperbolic discounting and environmental policy", *Environmental and Resources Economics*, Vol. 46, pp. 189-206
SP31	Hill, A. (2018), "Why nudges coerce: experimental evidence on the architecture of regulation", *Science and Engineering Ethics*, Vol. 24, pp. 1279-1295
SP32	Junghans, A. F., Cheung, T. T. and Ridder, D. D. D. (2015), "Under consumers' scrutiny-an investigation into consumers' attitudes and concerns about nudging in the realm of health behavior", *BMC Public Health*, Vol. 15, p. 336
SP33	Kaplan, B. A., Gelino, B. W. and Reed, D. D. A. (2018), "Behavioral economic approach to green consumerism: demand for reusable shopping bags", *Behavior and Social Issues,* Vol. 27, pp. 20-30
SP34	Kooreman, P. and Prast, H. (2010), "What does behavioral economics mean for policy? Challenges to savings and health policies in the Netherlands", *De Economist*, Vol. 158, pp. 101-122
SP35	Krishna, A. and Slemrod, J. (2003), "Behavioral public finance: tax design as price presentation", *International Tax and Public Finance*, Vol. 10, pp. 189-203
SP36	Lakerveld, J., Mackenbach, J. D., de Boer, F., Brandhorst, B., Broerse, J. E. W. and de Bruijn, G. J. (2018), "Improving cardiometabolic health through nudging dietary behaviours and physical activity in low SES adults: design of the Supreme Nudge project", *BMC Public Health*, Vol. 18, p. 899
SP37	Leeman, J., Myers, A. E., Ribis, K. M. and Ammerman, A. S. (2015), "Disseminating policy and environmental change interventions: insights from obesity prevention and tobacco control", *International Journal of Behavioral Medicine*, Vol. 22, pp. 301-311
SP38	Lepenies, R. and Małecka, M. (2015), "The institutional consequences of nudging – nudges, politics, and the law", *Review of Philosophy and Psychology*, Vol. 6, pp. 427-437
SP39	MacKay, K. and Quigley, M. (2018), "Exacerbating inequalities? Health policy and the behavioural sciences", *Health Care Anal*, Vol. 26, pp. 380-397
SP40	McMahon, J. (2015), "Behavioral economics as neoliberalism: producing and governing homo economicus", *Contemporary Political Theory*, Vol. 14, pp. 137-158
SP41	Metcalfe, J. J., Ellison, B., Hamdi, N., Richardson, R. and Prescott, M. P. (2020), "A systematic review of school meal nudge interventions to improve youth food behaviors", *International Journal of Behavioral Nutrition and Physical Activity*, Vol. 17, pp. 1-19
SP42	Nagatsu, M. (2015), "Social nudges: their mechanisms and justification", *Review of Philosophy and Psychology*, Vol. 6, pp. 481-494
SP43	Nakamura, R., Suhrcke, M. and Zizzo, J. D. (2017), "A triple test for behavioral economics models and public health policy", *Theory and Decision*, Vol. 83, pp. 513-533
SP44	Namazu, M., Zhao, J. and Dowlatabadi, H. (2018), "Nudging for responsible carsharing: using behavioral economics to change transportation behavior", *Transportation*, Vol. 45, pp. 105-119
SP45	Peeters, R. and Schuilenburg, M. (2017), "The birth of mindpolitics: understanding nudging in public health policy", *Social Theory and Health*, Vol. 15, pp. 138-159
SP46	Quigley, M. (2014), "Are health nudges coercive?" *Monash Bioethics Review*, Vol. 32, pp. 141-158

(*continued*)

Table A1.

Paper code	Details: author/s, year, title, journal, volume, number, pages
SP47	Rischkowsky, F. and Döring, T. (2008), "Consumer policy in a market economy: considerations from the perspective of the economics of information, the new institutional economics as well as behavioural economics", *Journal of Consumer Policy*, Vol. 31, pp. 285-313
SP48	Schnellenbach, J. (2019), "Evolving hierarchical preferences and behavioral economic policies", *Public Choice*, Vol. 178, pp. 31-52
SP49	Schnellenbach, J. and Schubert, C. (2019), "A note on the behavioral political economy of innovation policy", *Journal of Evolutionary Economics*, Vol. 29, pp. 1399-1414
SP50	Shogren, J. F., Parkhurst, G. M. and Banerjee, P. (2010), "Two cheers and a qualm for behavioral environmental economics", *Environmental and Resource Economics*, Vol. 46, pp. 235-247
SP51	Smith, V. K. and Moore, E. M. (2010), "Behavioral economics and benefit cost analysis", *Environmental and Resource Economics*, Vol. 46, pp. 217-234
SP52	Tagliabue, M. and Sandaker, I. (2019), "Societal well-being: embedding nudges in sustainable cultural practices", *Behavior and Social Issues*, Vol. 28, pp. 99-113
SP53	Viscusi, W. K. (2019), "Efficiency criteria for nudges and norms", *Public Choice*, doi: https://doi.org/10.1007/s11127-019-00712-5
SP54	Vlaev, I., King, D., Darzi, A. and Dolan, P. (2019), "Changing health behaviors using financial incentives: a review from behavioral economics", *BMC Public Health*, Vol. 19, p. 1059
SP55	Weijers, R. J., Koning, B. B. and Paas, F. (2020), "Nudging in education: from theory towards guidelines for successful implementation", *European Journal of Psychology of Education*, doi: https://doi.org/10.1007/s10212-020-00495-0
SP56	Yoong, S. L., Hall, A., Stacey, F., Grady, A., Sutherland, R., Wyse, R., Anderson, A., Nathan, N. and Wolfenden, L. (2020), "Nudge strategies to improve healthcare providers' implementation of evidence based guidelines, policies and practices: a systematic review of trials included within Cochrane systematic reviews", *Implementation Science*, Vol. 15, p. 50
W1	Baldwin, R. (2014), "From regulation to behaviour change: giving nudge the third degree", *The Modern Law Review*, Vol. 77 No. 6, pp. 831-857
W2	Battaglio, R. P., Belardinelli, P., Bellé, N. and Cantarelli, P. (2018), "Behavioral public administration ad fontes: a synthesis of research on bounded rationality, cognitive biases, and nudging in public organizations", *Public Administration Review*, Vol. 79 No. 3, pp. 304-320
W3	Byerly, H., Balmford, A., Ferraro, P. J., Wagner, C. H., Palchak, E., Polasky, S., Ricketts, T. H., Schwartz, A. R. and Fisher, B. (2018), "Nudging pro-environmental behavior: evidence and opportunities", *Frontiers in Ecology and the Environment*, Vol. 16 No. 3, pp. 159-168
W4	Cai, C. W. (2018), "Nudging the financial market? A review of the nudge theory", *Accounting and Finance*, Vol. 60 No. 4, pp. 3341-3365
W5	Clark, R. L., Maki, J. A. and Morrill, M. S. (2014), "Can simple informational nudges increase employee participation in a 401(k) plan?" *Southern Economic Journal*, Vol. 80 No. 3, pp. 677-701
W6	Davidson, K. A., Kropp, J. D., Mullally, C. and Rahman, M. W. (2020), "Can simple nudges and workshops improve diet quality? Evidence from a randomized trial in Bangladesh", *American Journal of Agricultural Economics*, Vol. 103 No. 10, pp. 1-22
W7	French, R. and Oreopoulos, P. (2017), "Applying behavioural economics to public policy in Canada", *Canadian Journal of Economics*, Vol. 50 No. 3, pp. 599-635
W8	Galizzi, M. M. (2014), What is really behavioral in behavioral health policy? And does it work? *Applied Economic Perspectives and Policy*, Vol. 36 No. 1, pp. 25-60
W9	Gopalan, M. and Pirog, M. A. (2017), "Applying behavioral insights in policy analysis: recent trends in the United States", *The Policy Studies Journal*, Vol. 45 No. S1, pp. 82-114
W10	Gowayed, H. (2018), "The unnecessary nudge: education and poverty policy in a cairo slum", *Sociological Forum*, Vol. 33 No. 2, pp. 1-23
W11	Guthrie, J., Mancino, L. and Lin, C. T. J. (2015), "Nudging consumers toward better food choices: policy approaches to changing food consumption behaviors", *Psychology and Marketing*, Vol. 32 No. 5, pp. 501-511

Table A1. (*continued*)

Paper code	Details: author/s, year, title, journal, volume, number, pages
W12	Meyer, K. and Rosinger, K. O. (2019), "Applying behavioral insights to improve postsecondary education outcomes: a review of Obama administration efforts and next steps under the trump administration", *Journal of Policy Analysis and Management*, Vol. 38 No. 2, pp. 481-499
W13	Oliver, A. (2015), "Nudging, shoving, and budging: behavioural economic-informed policy", *Public Administration*, Vol. 93 No. 3, pp. 700-714
W14	Oliver, A. (2019), "Towards a new political economy of behavioral public policy", *Public Administration Review*, Vol. 79 No. 6, pp. 917-924
W15	Petticrew, M., Maani, N., Pettigrew, L., Rutter, H. and Schalkwyk, M. C. V. (2020), "Dark nudges and sludge in big alcohol: behavioral economics, cognitive biases, and alcohol industry corporate social responsibility", *The Milbank Quarterly*, Vol. 98 No. 4, pp. 1290-1328
W16	Pogarsky, G. and Herman, S. (2019), "Nudging and the choice architecture of offending decisions", *Criminology and Public Policy*, Vol. 18, pp. 823-839
W17	Rachlin, H. (2015), "Choice architecture: a review of why nudge: the politics of libertarian paternalism", *Journal of the Experimental Analysis of Behavior*, Vol. 104 No. 2, pp. 198-203
W18	Tummers, L. (2019), "Public policy and behavior change", *Public Administration Review*, Vol. 79 No. 6, pp. 925-930
W19	Weimer, D. L. (2020), "When are nudges desirable? Benefit validity when preferences are not consistently revealed", *Public Administration Review*, Vol. 80 No. 1, pp. 118-126

Table A1.

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